PROJECT MANAGEMENT

Presented by

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Content

Financial Analysis

- Costs
- Break Even Analysis
- Net Present Value (NPV)
- Internal Rate of Return (IRR)
- Sensitivity analysis
- Preparation of balance sheets

Financial Analysis

Project financial management involves the processes of

i. Estimating
ii. Budgeting
iii. Tracking
iv. Controlling

the financial aspects of a project throughout its lifecycle.

This includes managing costs, revenues, and profits to ensure that the project remains financially viable and meets its objectives.

Key Components of Project Financial Management

Budgeting: Establishing a clear budget is essential. This involves **forecasting costs**, **allocating resources**, and **setting financial limits** for various project activities. Effective budgeting helps **prevent overspending** and ensures that **funds are available** when needed.

Cost Control: Monitoring and controlling project costs is vital to maintain financial discipline. This includes **tracking actual expenses against the budget, identifying variances, and implementing corrective actions** when necessary.

Revenue Management: Understanding the **sources of revenue** generated by the project is important. This can include **service fees, product sales, or other income streams**. Proper revenue management ensures that the project remains profitable.

Key Components of Project Financial Management

Financial Reporting: Regular financial reporting provides stakeholders with insights into the project's financial health. This includes **updates on budget status, cost overruns, and overall financial performance**.

Risk Management: Financial management also involves **assessing financial risks** associated with the project. This includes **identifying potential financial pitfalls and developing strategies to mitigate them**.

Costs

Direct cost	Direct costs are those **directly linked** to doing the work of the project. For example, this could include hiring specialised contractors, buying software licences or commissioning your new building.
Indirect cost	These costs are **not specifically linked to your project** but are the **cost of doing business overall**. Examples are heating, lighting, office space rental (unless your project gets its own offices hired specially), stocking the communal coffee machine and so on.
Fixed cost	Fixed costs are everything that is a **one-off charge**. These fees are not linked to how long your project goes on for. So if you need to pay for one-time advertising to secure a specialist software engineer, or you are paying for a day of Agile consultancy to help you start the project up the best way, those are fixed costs.
Variable cost	These are the **opposite of fixed costs - charges that change with the length of your project.** It's more expensive to pay staff salaries over a 12 month project than a 6 month one. Machine hire over 8 weeks is more than for 3 weeks.
Sunk cost	These are costs that have **already been incurred**. They could be made up of any of the types of cost above but the point is that they have happened. The money has gone. These costs are often forgotten in business cases, but they are essential to know about.

Break-Even Analysis

Refers to identifying the **point where the revenue of the company starts exceeding its total cost** i.e., the point when the project or company under consideration will **start generating the profits** by the way of studying the relationship between the revenue of the company, its fixed cost, and the variable cost.

There are **two approaches** to calculate the break-even point:

i. Quantity - break-even quantity

Divide the fixed cost by contribution per unit i.e.

Break-Even Point (Qty) = Total Fixed Cost / Contribution per Unit

Contribution per Unit = Selling Price per Unit - Variable Cost per Unit

ii. Sales - break-even sales

Divide the fixed cost by the contribution-to-sales ratio or profit-volume ratio i.e.

Break-Even Sales (Rs) = Total Fixed Cost / Contribution Margin Ratio

Contribution Margin Ratio = Contribution per Unit / Selling Price per Unit

Break-Even Analysis

- For conducting a mining project's break-even analysis, **operational expenses (OPEX)** needs to be known.
- When the OPEX is known, the mineral's cut off grade can be calculated, which is **the break even grade, below which it is not economically viable to mine the ore**.
- When the data about cost price per tonne (OPEX) cannot be found then a best estimate shall be made.
- **Calculating the cut-off grade** helps mining companies decide whether a deposit is worth pursuing and aids in the optimization of mine planning and resource allocation.
- It is important to note that *the* ***cut-off grade can change over time as market prices fluctuate, mining costs evolve, and new technologies are developed***.

Mining Project Cash Flow

- Basic mining-project cash-flow structure

For each year t:

Free Cash Flow (FCF_t) = $Revenue_t - OPEX_t - Tax_t - CAPEX_t - \Delta WC_t$ + Salvage

where:

Revenue = ROM/coal/ore production × selling price

OPEX = mining + processing + transportation + administration, etc.

CAPEX = initial and sustaining capital expenditure

Tax = income tax and other applicable project taxes/levies

ΔWC = change in working capital

Salvage = residual value of equipment/assets at project closure

Net Present Value (NPV)

- NPV is the widely accepted tool for **measuring the value of a mine project.**
- In the dynamic field of mining, where financial stakes are high, NPV emerges as the most prevalent tool for guiding significant **financial decisions**.
- **NPV is the present value of all future project cash flows minus the initial investment.**
- **NPV represents the total value of an investment's future cash flows, brought back to their present value.**
- Calculates the difference between the present value of cash inflows and outflows over a project's lifespan.
- It reflects the profitability of a project after accounting for the time value of money.
  - **Positive NPV - Profitable investment.**
  - **Negative NPV - Potential losses.**

Net Present Value (NPV)

To calculate NPV:

i. Estimate the annual net cash flows (revenues minus costs) of an investment.

ii. These are then adjusted using a predetermined discounting rate to reflect their present value.

iii. The NPV is the sum of these discounted cash flows.

- **It calculates the current value of the cash flows**

at the required rate of return of a project, compared

to the initial investment.

$$\text{NPV} = \frac{R_t}{(1+i)^t}$$

NPV = Net Present Value, R(t)= Net cash flow at time t, i = discount rate, t= timeof cash flow

Calculating NPV

$$NPV = \sum_{t=0}^{n} \frac{FCF_t}{(1+r)^t}$$

r = discount rate; t = year; n = project life.

NPV > 0: attractive at the selected discount rate.

NPV = 0: break-even.

NPV < 0: unattractive.

Because the initial investment occurs at t=0:

$$NPV = -CAPEX_0 + \frac{FCF_1}{(1+r)} + \frac{FCF_2}{(1+r)^2} + \cdots + \frac{FCF_n}{(1+r)^n}$$

Internal Rate of Return (IRR)

- **IRR is a rate of return on an investment.**
- The IRR of an investment is the interest rate that gives it a NPV of 0, or where the sum of discounted cash flow is equal to the investment.
- Represents the **discount rate** that makes the **Project NPV = 0**.
- The IRR is calculated by trial and error.
- IRR is the best method to evaluate the economic side of an investment, because it allows a good comparison with other investment projects and financial alternatives (bank account, stocks, real estate) (Gessinger, 2009).

$$0 = \sum_{t=0}^{n} \frac{FCF_t}{(1 + IRR)^t}$$

Sensitivity Analysis

- **Sensitivity analysis** involves assessing how changes in various assumptions can impact the financial performance of a mining project.
- This analysis helps stakeholders understand the project's resilience to different scenarios and uncertainties.

[Mining Sensitivity Analysis: Key Parameters, Base Case Scenario, and Strategies | AllMinings]

Steps involved in conducting a sensitivity analysis for mining operations economics:

Define Key Parameters

Establish Base Case Scenario

Vary Parameters

Assess Financial Metrics

Plot Results

Determine Sensitivity Ranges

Perform Scenario Analysis

Evaluate Risk and Uncertainty

Identify Risk Mitigation Strategies

Document and Communicate Findings

Sensitivity Analysis

- Commodity price
- Operating cost and CAPEX
- Grade, recovery and production rate
- Mine life and discount rate
- Open-pit: stripping ratio
- Underground: dilution, recovery and mining method
- Use a tornado chart to present NPV sensitivity.

Preparation of Balance Sheets

Preparing a projected balance sheet for a project involves creating a financial statement that shows the estimated assets, liabilities, and equity of the project at a specific point in the future.

1. Gather Data and Assumptions
2. Choose the Projection Period
3. Start with the Beginning Balance
4. Estimate Projected Assets
5. Estimate Projected Liabilities
6. Calculate Equity
7. Add or Deduct Equity Changes
8. Complete the Balance Sheet
9. Perform Sensitivity Analysis
10. Validate and Review
11. Presentation and Reporting
12. Monitor and Update

[Prepare projected balance sheets for a project - Noteswa]

References

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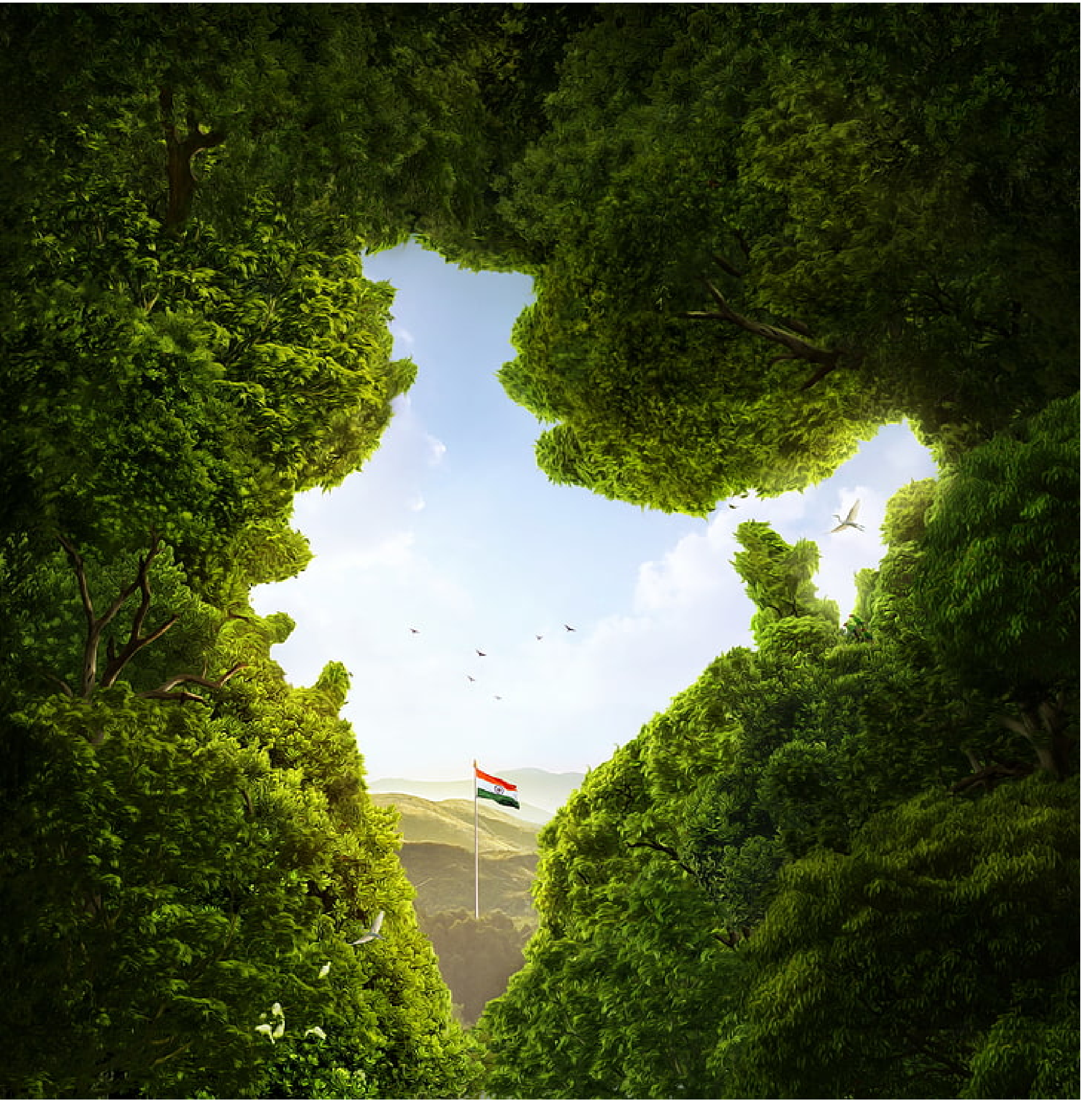

Thank You

Any Queries ???

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